

Explore the Data from Different Sources

Aim:

To Write a program to explore the Data from Different Sources.

ALGORITHM:

STEP: 1 $\Rightarrow$ Open Tableau Workspace and Minimize the Screen. Then create a table in M.S Excel with the following fields: (Roll.No, Name, Class, Department, Mark, State, Company)

STEP: 2 $\Rightarrow$ Enter 15 records, and Save the file.

STEP: 3 $\Rightarrow$ Open Tableau $\rightarrow$ Select M.S. Excel from the connect panel $\rightarrow$ Locate and open the saved excel file from your Desktop.

STEP: 4 $\Rightarrow$ Click on select 1 from the bottom Tab to begin Visualization.

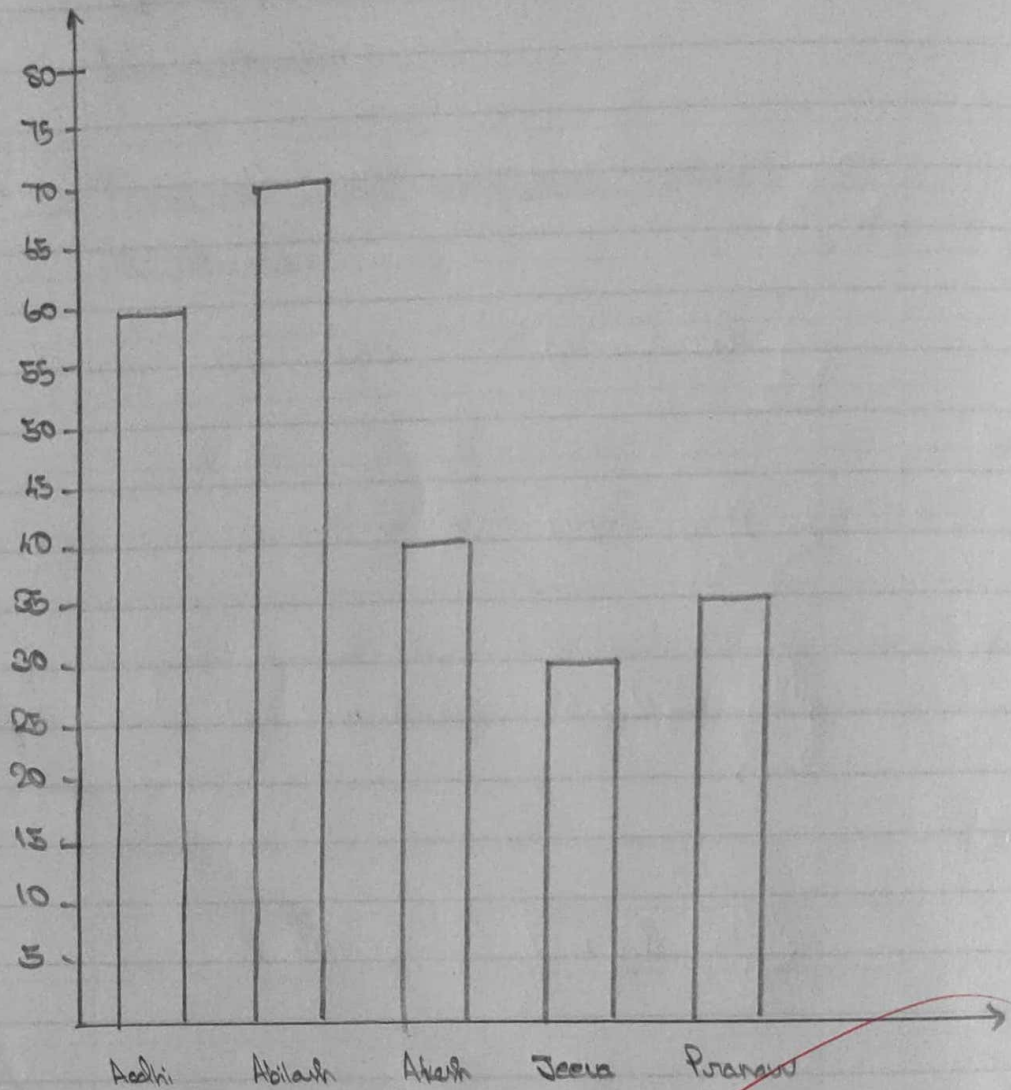
STEP: 5 $\Rightarrow$ Drag and Drop the fields from The Data To different rows and columns. Use "Show me" to Visualize the Data

STEP: 6 $\Rightarrow$ you can also Customize the Visual by choosing colors or styles to Match your references.

RESULT:

Thus the Output is executed Successfully.

Output:



Aim:

To Write a Program to Organize and Filter Datas using Tableau.

ALGORITHM:

STEP: 1 $\Rightarrow$ Connect to Data, open Tableau $\rightarrow$ click connect $\rightarrow$ choose excel.

STEP: 2 $\Rightarrow$ Organize data, drag and relevant dimensions and measures to rows and columns, Sort data by clicking the Sort.

STEP: 3 $\Rightarrow$ Apply Filter, drag the field you want to filter, to the filter shelf with condition.

STEP: 4 $\Rightarrow$ Create Visualization, choose a chart from "Show me" panel, customize columns, labels and title as needed.

STEP: 5 $\Rightarrow$ View and Share, use dashboard to Customize Multiple Views and export or publish to Tableau public or server.

RESULT:

Thus the output is executed Successfully.

Output:

Sheet 1

Prod	Sales Amt.	
Car	40 crore	Lucknow 
Trilight	55 crore	New Delhi 
bike	36 crore	Hydrabad 
Scooter	42 Crore	TamilNadu 
Auto	10 crore	Punjab 
Cycle	10 Crore	TamilNadu 

Exhibit analytics using Data Aggregation.

Aim:

To Write a Program to Exhibit analytics using Data Aggregation.

ALGORITHM:

STEP: 1 $\Rightarrow$ Open MS Excel and create a dataset with measurable field including: Transaction ID, Cust name, Sales, Profit, region, Prod. Name and Prod. category, 5 rows.

STEP: 2 $\Rightarrow$ Open Tableau and connect to data. Then click worksheet to start building, Aggregate Data for Analytics.

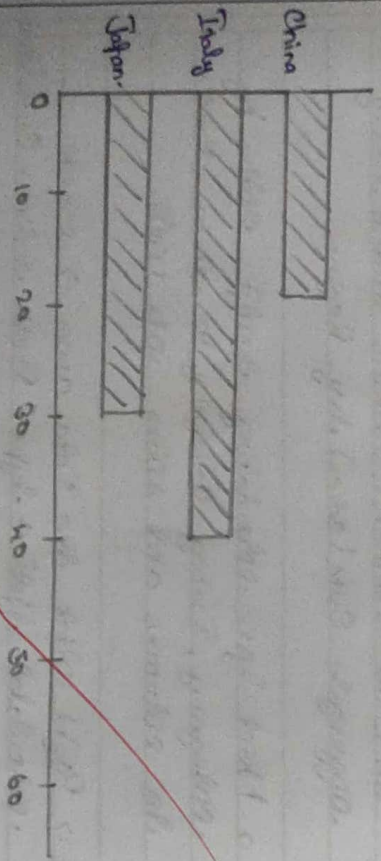
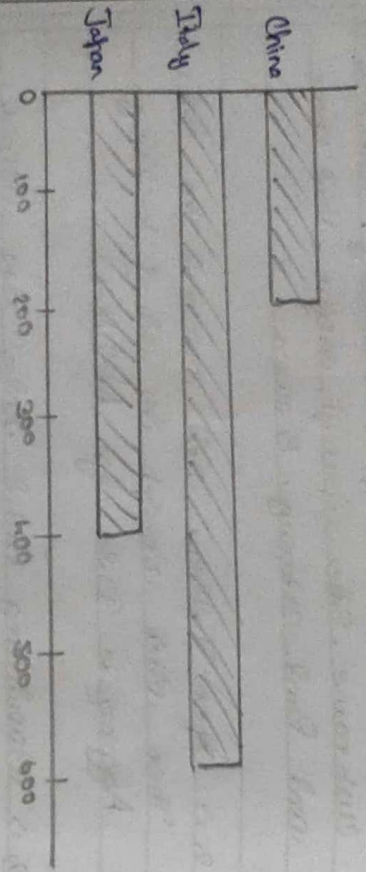
STEP: 3 $\Rightarrow$ Calculating total sales by region, for this drag region to rows and Sales to column. Tableau automatically aggregate $\text{Sum}(\text{Sales})$ by Region.

STEP: 4 $\Rightarrow$ Next for calculating profit ratio by category, Drag Category to rows, Profit to columns and sales also well.

STEP: 5 $\Rightarrow$ Right click the Data area $\rightarrow$ create calculated field $\rightarrow$ type " $\text{Profit ratio} = \text{Sum}(\text{Profit}) / \text{Sum}(\text{Sales})$ " and drag the field to chart to impact.

Output:

Region	Product
China	Shirt
Italy	Foot
Japan	Shoe
Spain	Bag



Ex. No.

Date :

Result:

Thus the output is recorded successfully

Analyse the proportional relationship among the data.

Aim:

To write a program to analyse the proportional relationship among the Data.

ALGORITHM:

STEP: 1 $\Rightarrow$ Import dataset into Tableau.

STEP: 2 $\Rightarrow$ Identify a categorical field and a Numerical Measures.

STEP: 3 $\Rightarrow$ Choose a chart that show proportion (Piechart, Tree Map, Stacked bar)

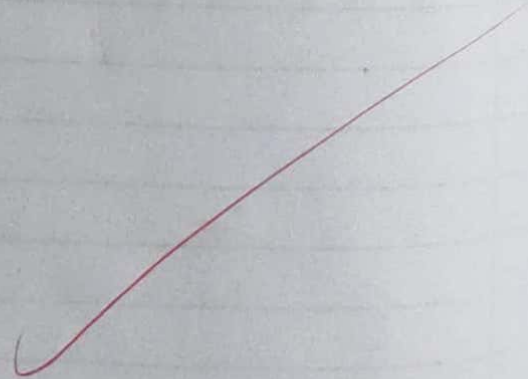
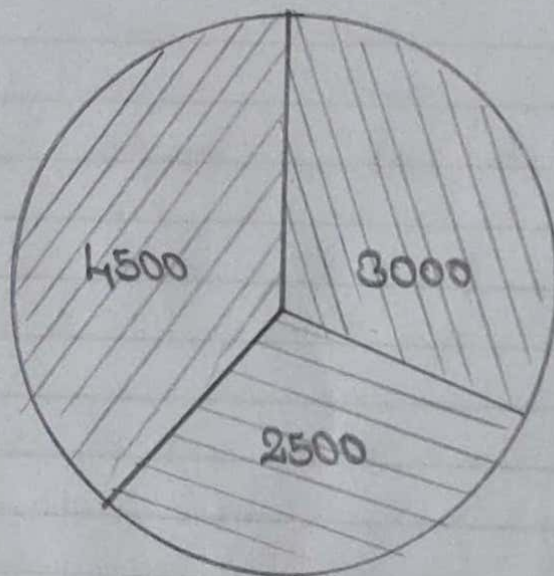
STEP: 4 $\Rightarrow$ Assign the categorical field to color / label and the Numerical field to Size / angle.

STEP: 5 $\Rightarrow$ Interpret proportional relationship Visually.

RESULT:

Thus the output is recorded Successfully.

Answer:



Visualize Measure against Dimensions

Aim:

To Write a program to Visualize a Measures Dimensions in Tableau Using Maps.

ALGORITHM:

STEP:1 $\Rightarrow$ Open Tableau and load your dataset (Country and sales as field)

STEP:2 $\Rightarrow$ Use geographical Dimensions like (city, region) and measures (like Sales, Profit)

STEP:3 $\Rightarrow$ Drag a geographical field (eg. city) to the View - Tableau auto-generates a Map.

STEP:4 $\Rightarrow$ Drag Measure, country to rows and Sales to colors, size and detail to Visualize Data.

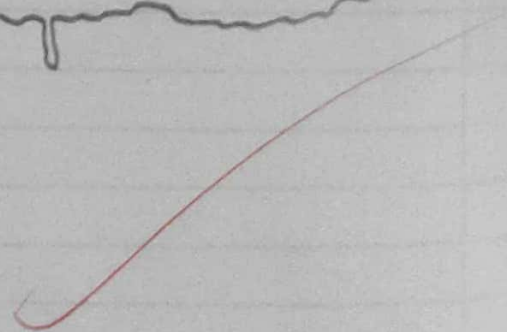
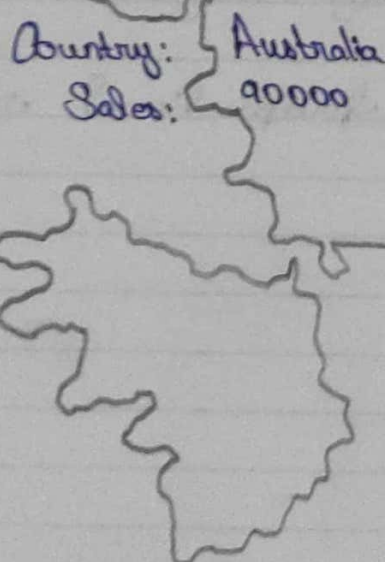
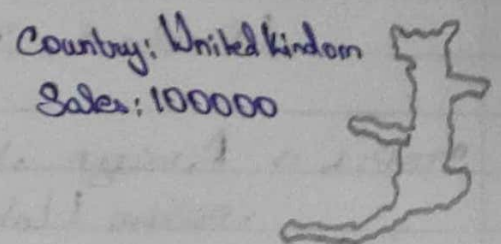
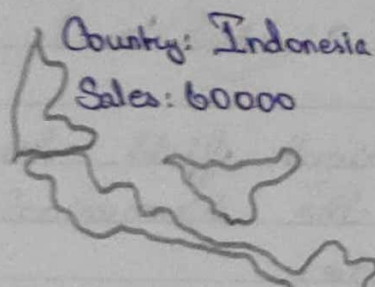
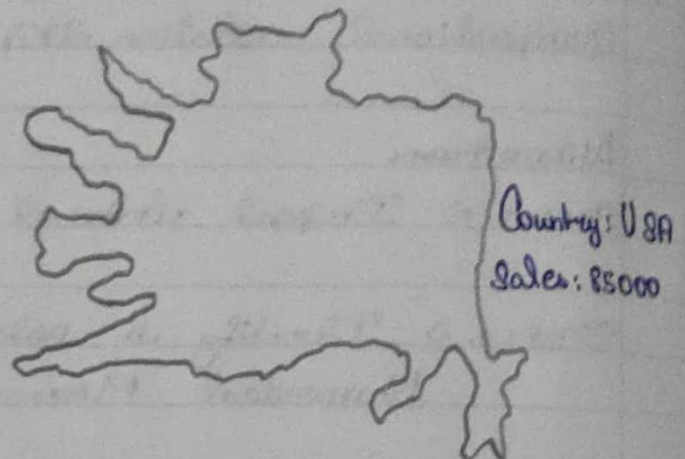
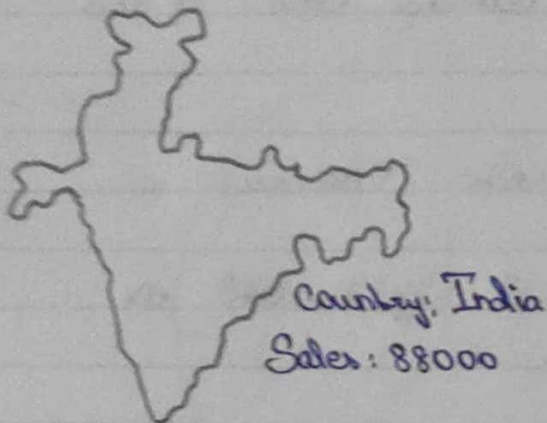
STEP:5 $\Rightarrow$ Under "Show me" panel changes the Visualization format to "Map".

STEP:6 $\Rightarrow$ Add filters and edit toolpits to Show relevant information when hovering.

RESULT:

Thus the output is executed successfully.

Output:



Aim:

To Write a program to recognize your Location Data and automatically assign geographical roles.

Algorithm:

STEP: 1 $\Rightarrow$ Open Tableau and load your Datasets into Worksheet and double-click a geographical field (eg. country) to generate Map.

STEP: 2 $\Rightarrow$ From the map layer pane, click Map $\rightarrow$ Map layers.

STEP: 3 $\Rightarrow$ We can also customize the Map.

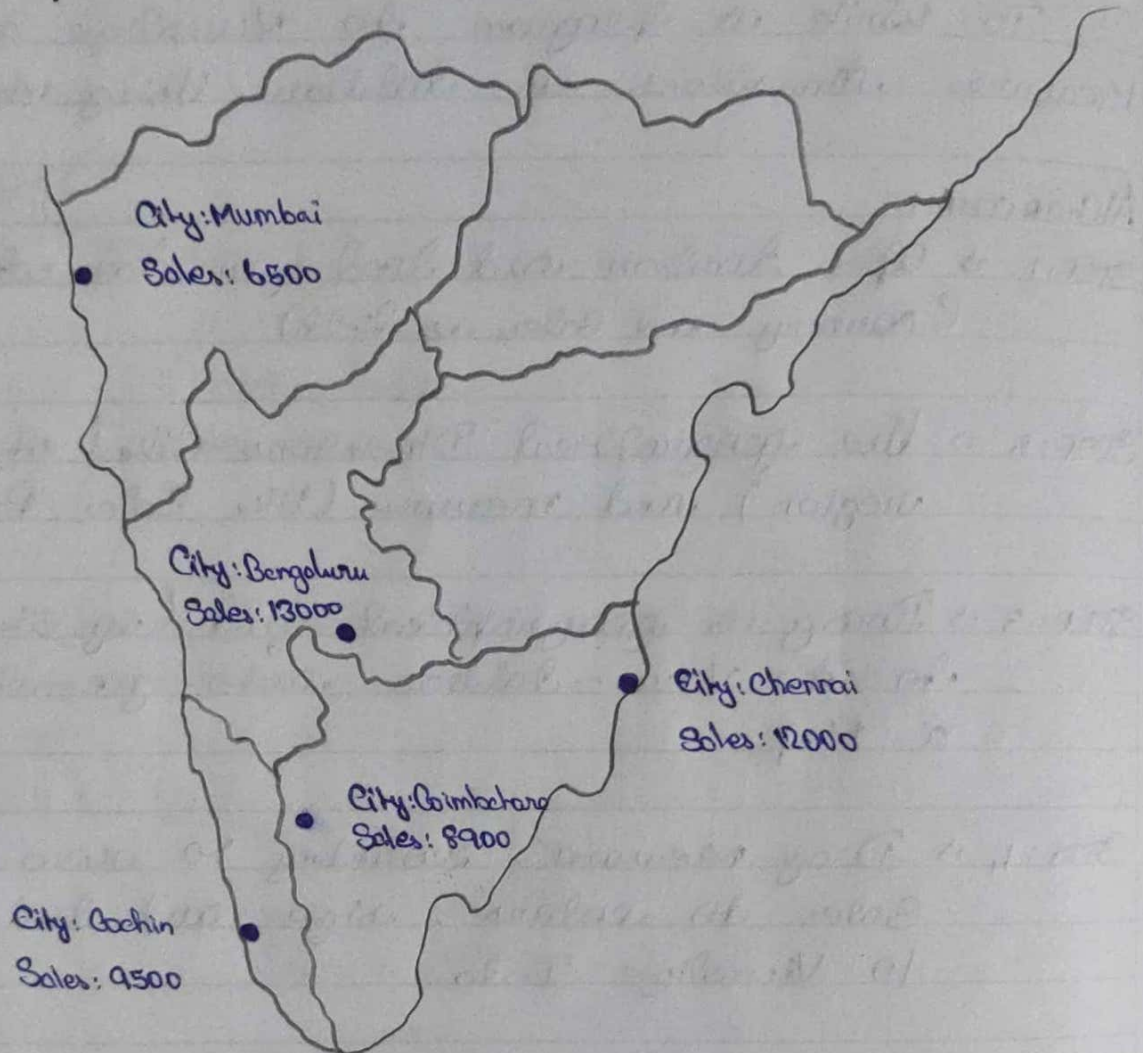
STEP: 4 $\Rightarrow$ Drag state to details and city also to detail. Now all the cities will be Plotted.

STEP: 5 $\Rightarrow$ Now we can drag Sales to size, colors so that the cities will be sized as bubbles and colored according to the Sales amount.

Result:

Thus the output is executed successfully.

Output:



Compare sets of Data Using
Plots

Aim:

To write a program to compare a sets of Data using Plots.

ALGORITHM:

STEP: 1 $\Rightarrow$ Create region, Sales, Profit fields in excel sheet.

STEP: 2 $\Rightarrow$ Drag region to column sheet, Sales to rows and Profit to rows.

STEP: 3 $\Rightarrow$ Create a side by side chart bar.

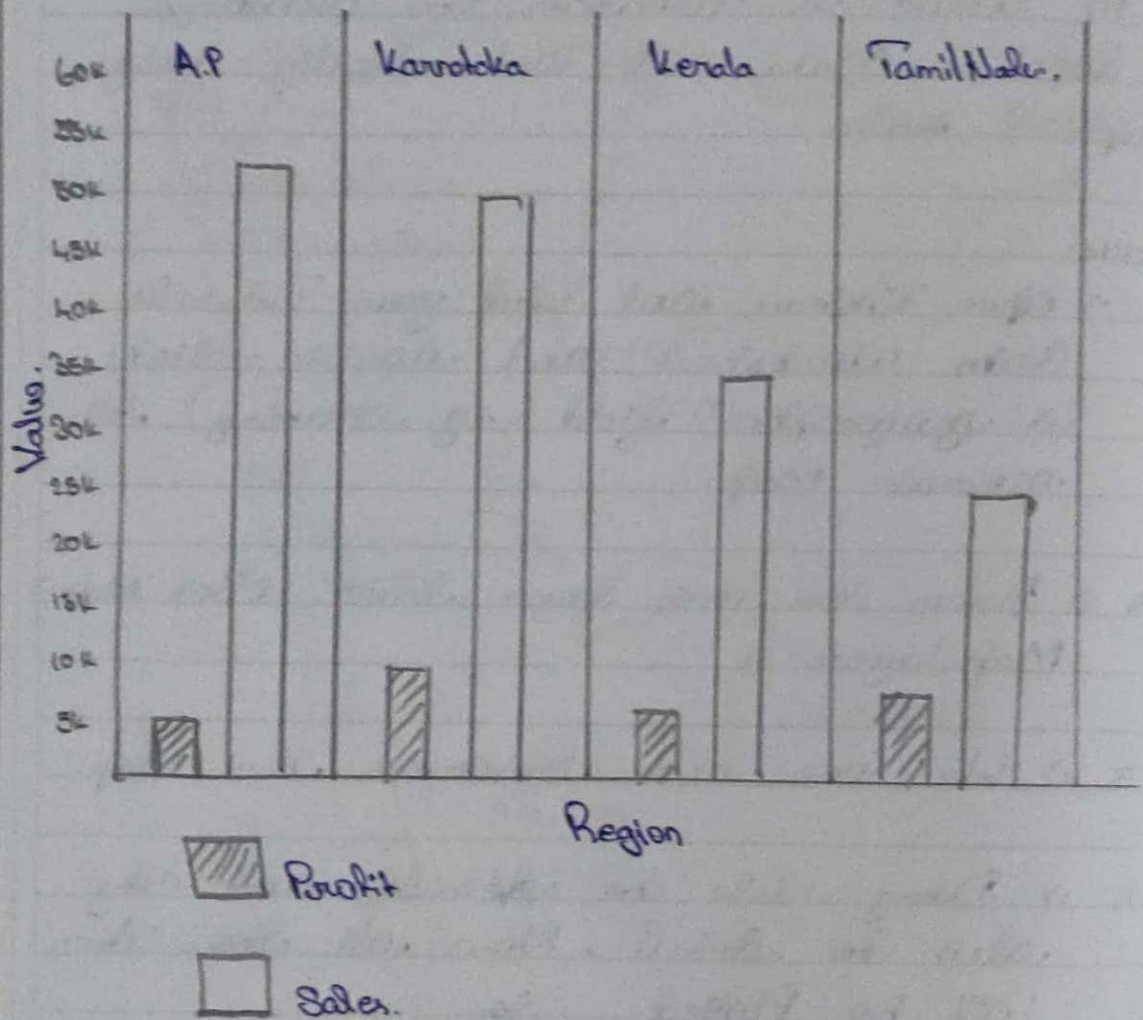
STEP: 4 $\Rightarrow$ Customize colors for described fields like sales and Profit.

STEP: 5 $\Rightarrow$ Filter the Data based on specific region. drag the regions field to filter sheet and select the regions to the filters.

RESULT:

Thus the output is executed successfully.

Output:



Build a calculated field for any data source.

AIM:

To write a program to Build a calculated field for any data source.

ALGORITHM:

STEP: 1 $\Rightarrow$ Open Tableau public and connect to Dataset.

STEP: 2 $\Rightarrow$ Now right click anywhere on Sheet 1

STEP: 3 $\Rightarrow$ Select create calculated field, A Dialog box will appear.

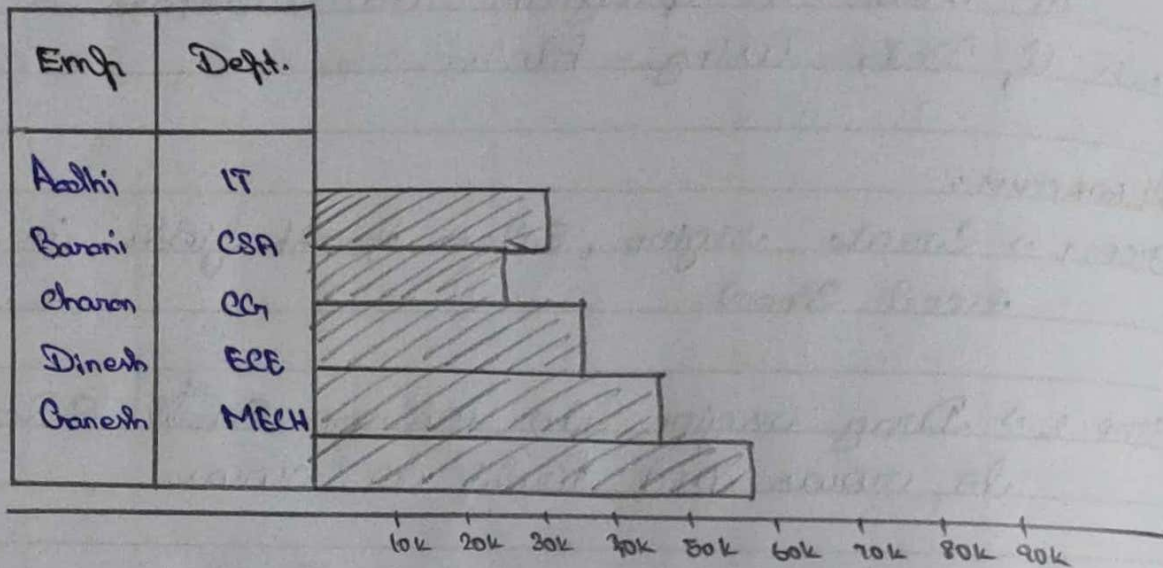
STEP: 4 $\Rightarrow$ In Dialog box, give Name as Net Salary, inside The show type $[basic\ Salary] + [Bonus] - [Deduction]$

STEP: 5 $\Rightarrow$ Now output appeared for this Net Salary.

RESULT:

Thus the output is executed successfully.

Output:



Build a effective Dashboard of your company Profile.

Aim:

To Write a program to Build an effective Dashboard of your company Profile.

ALGORITHM:

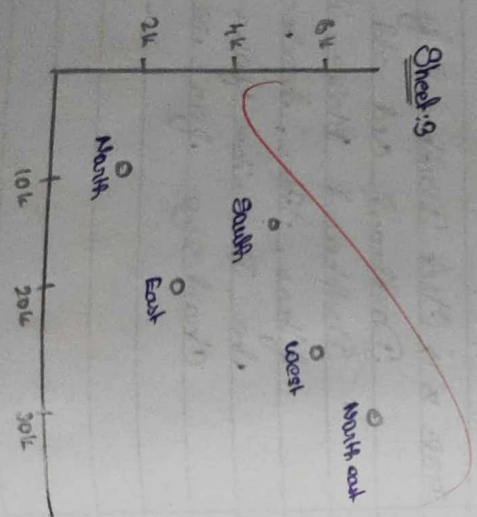
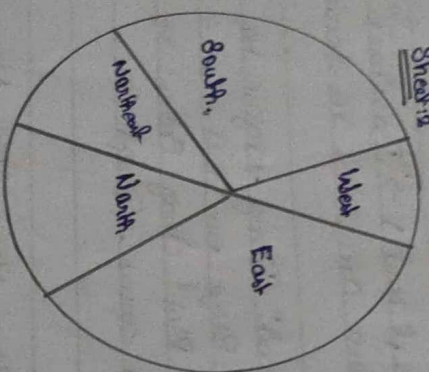
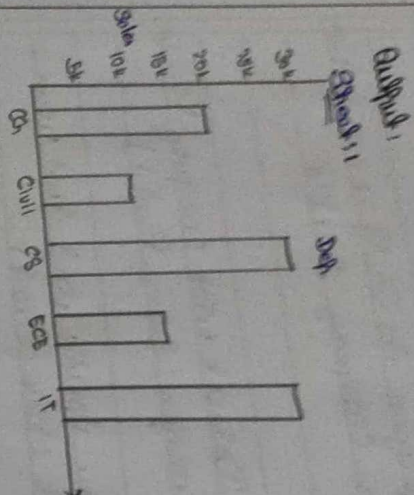
STEP: 1 $\Rightarrow$ Create a Dataset with company Profile.
[Emp, dept, region, Sales, Profit]

STEP: 2 $\Rightarrow$ Open Tableau public and load your Dataset and click sheet1

STEP: 3 $\Rightarrow$ Create 3 Charts, Chart 1 $\rightarrow$ "Bar chart"
Drag dept to column and Sales to rows

STEP: 4 $\Rightarrow$ Chart 2 $\rightarrow$ "Pie chart" Drag region to
color Profit to Size or label.
Chart 3 $\rightarrow$ "Scatter Plot" drag Sales to
column, Profit to rows, Dept to
color.

STEP: 5 $\Rightarrow$ Click Dashboard from Menu, Click New
Dashboard and add your sheet to
Dashboard Make it interactive by adding
filters, titles, toolboxes to Datasets
to Describe your Dashboard Adjust
chart size for readability.



Ex. No. _____
Date: _____

Result:

Thus The output is described
Successfully.

Describe Stories Using Different Dashboard and Worksheet.

Ans:

To Write a program to Describe Stories Using Different Dashboard and Worksheet.

ALGORITHM:

STEP: 1 $\Rightarrow$ Open Tableau public and load your Datasets

STEP: 2 $\Rightarrow$ Create 3 types of chart

$\Rightarrow$ bar chart

$\Rightarrow$ Pie chart

$\Rightarrow$ Scatter Plot

STEP: 3 $\Rightarrow$ Follow the Same Steps for the 3 charts from previous one.

STEP: 4 $\Rightarrow$ Create Dashboard 1 by clicking New Dashboard and Dragging the sheet.

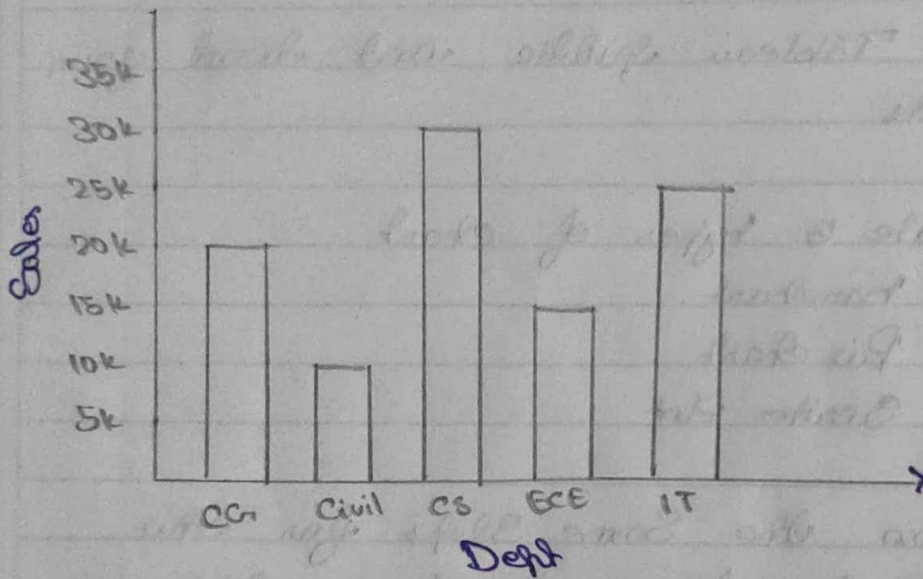
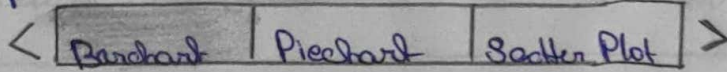
STEP: 5 $\Rightarrow$ Create Dashboard 2 by the Same Method as Above.

STEP: 6 $\Rightarrow$ Now Create a Story by Selecting from the Menu and drag the Dashboard into the field.

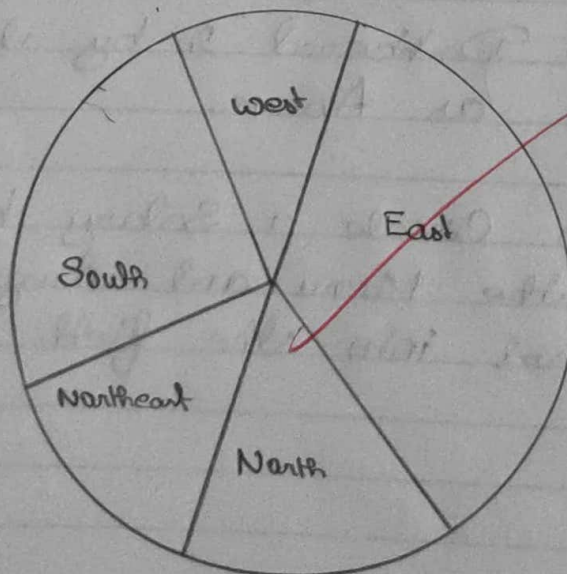
Output:

Story 1

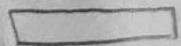
Sheet 1



Sheet 2



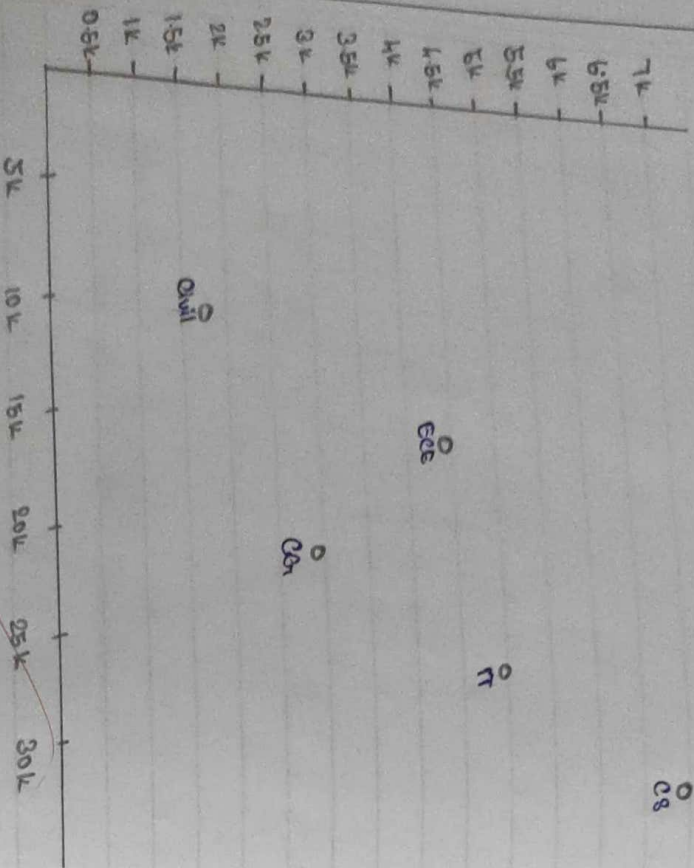
Profit



21,300

Sheet 3

Bar chart Pie chart Scatter Plot



Ex No.

Date :

Page No.

Result:

Thus the output is recorded successfully.

Forecast and Explore instant analysis

Aim:

To Write a program to forecast and Explore instant data analysis.

ALGORITHM:

STEP: 1 $\Rightarrow$ Create a timeseries dataset "Month
Year, sales"

STEP: 2 $\Rightarrow$ Open Tableau and load your dataset
and click sheet 1

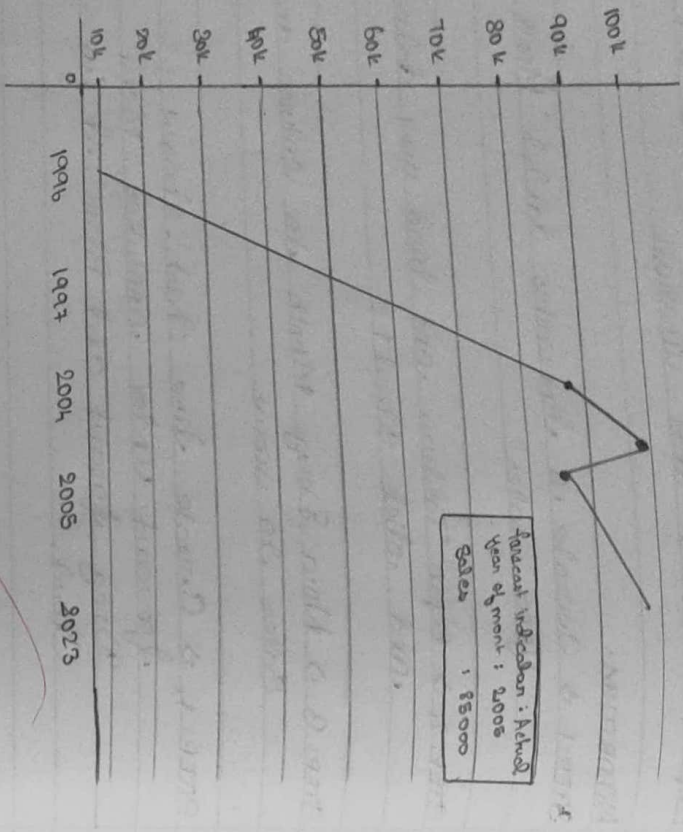
STEP: 3 $\Rightarrow$ Now Drag Month to column and
Sales to rows.

STEP: 4 $\Rightarrow$ Create line chart, choose
forecast under analytics, Menu
Drag forecast and Place it in
Chart.

STEP: 5 $\Rightarrow$ Now right click in chart
forecast option, choose Month as
year as forecast length and
Select automatic Model.

STEP: 6 $\Rightarrow$ Right click sales quick table
Calculation Using Percentage
Difference (or) Moving average.

Output



Ex. No.
Date:

Page No.

Result: Thus the output is recorded successfully.